DSA ki Baithak

CODE BAITHAK

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Recap (Previous Lecture Connection):

* Number System
* Base (Radix)
* Any Base Addition
* Decimal to Any Base Conversion
* Any Base to Decimal Conversion

Any base (b) → (2) Decimal (des) (10) | Decimal (src) (10) → (1) Any base (b)

[divide → des_base
multiply → scr_base]

Any Base System to Any Base System :

Any Base System (src) → Decimal (10) → Any Base System (des)

[destination base → division

source base → multiplication] → when either of base is decimal (10)

1's Complement of Decimal

$n = 53 \longrightarrow 110101$ → bit

$110101 \rightarrow 001010$

$001010 \longrightarrow 10$

(bits flip)

$\begin{bmatrix} 1 \rightarrow 0 \\ 0 \rightarrow 1 \end{bmatrix}$

1. decimal → binary
2. flip
3. binary → decimal

1

$n \rightarrow \cancel{53}\ \cancel{26}\ \cancel{13}\ \cancel{6}\ \cancel{3}\ \cancel{1}\ 0$

$pow = \cancel{1}\ \cancel{2}\ \cancel{2^2}\ \cancel{2^3}\ \cancel{2^4}\ \cancel{2^5}\ 2^6$

$ans = 0+0+2+0+8+0+0$

2	53	1
2	26	0
2	13	1
2	6	0
2	3	1
2	1	1
	0	

rem = $\cancel{1}\ \cancel{0}\ \cancel{1}\ \cancel{0}$

$\downarrow\ \downarrow\ \downarrow\ \downarrow$

0 1 0 1

1 1

$\downarrow\ \downarrow$

0 0

ans = 0, pow = 1

```
while(n > 0) {
    int rem = n % 2
    n /= 2
    ans += (1-rem) * pow
    pow *= 2
}
```

ans $\rightarrow$ 2+8 = 10